



Figure 1: Passive attacks in the network environment

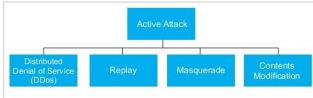


Figure 2: Active attacks in the network environment

as a HULK (HTTP unbearable load king) attack. In a HULK attack, the unbearable load is created at the HTTP service. A number of virtual connections are created and then fired at the website. If a HULK attack is used, the particular website gets a large number of connections from fake traffic and then the website hangs. That's why HULK is classified as a DDoS attack. HULK attacks are generally carried out using Python, PHP, Java or Perl scripts, which are easily available on assorted Web based repositories of source code.

As there are massive DDoS attacks on different websites and servers, it is mandatory for network administrators to adopt and implement mechanisms to cope with and mitigate such attacks.

Let's now discuss two free and open source tools that help to detect and repel such DDoS attacks.

DDoS Deflate

DDoS Deflate, an open source tool, is a powerful shell script to cope with DDoS attacks on servers. DDoS Deflate is dominant enough to push back and block DDoS attacks. At the base level, it makes use of the *netstat* command to identify and investigate the IP addresses that are creating connections with the server.

The following command is used to identify and list the connections created by all the IP addresses:

```
<workingdirectory>$ netstat -ntu | awk '{print $5}' | cut -d: -f1 | sort | uniq -c | sort -n
```

The features of DDoS Deflate include:

- Auto blocking of IP addresses
- Blacklisting and whitelisting of traffic and their sources
- Easy notification and management for network administrators
- Auto detection of rules associated with Iptables and advanced policy firewalls
- Ease in configuration

- Auto e-mail alerts
- Uses *tcpkill* to push back the unwanted and fake connections

Fire the following commands in a terminal to install DDoS Deflate:

```
<workingdirectory>$ cd /usr/local/src/
<workingdirectory>$ wget http://www.inetbase.com/scripts/ddos/install.sh
<workingdirectory>$ chmod 0700 install.sh
<workingdirectory>$ ./install.sh
```

The configuration file of DDoS Deflate can be edited as follows:

```
<workingdirectory>$ vi /usr/local/ddos/ddos.conf
```

or

```
<workingdirectory>$ gedit /usr/local/ddos/ddos.conf
```

Use the following command to start DDoS Deflate:

```
<workingdirectory>$ /usr/local/ddos/ddos.sh -c
```

The following code will uninstall DDOS Deflate:

```
<workingdirectory>$ wget http://www.inetbase.com/scripts/ddos/uninstall.ddos
<workingdirectory>$ chmod 0700 uninstall.ddos
<workingdirectory>$ ./uninstall.ddos
```

To view the 'Help' screen and all other options in DDoS Deflate, type:

```
<workingdirectory>$ ddos -help
```

To view whitelisted IP addresses, type:

```
<workingdirectory>$ ddos -I | -ignore-list
```

Use the following command to view banned or blacklisted IP addresses:

```
<workingdirectory>$ ddos -b | -bans-list
```

To initialise the daemon process for monitoring connections, type:

```
<workingdirectory>$ ddos -d | -start:
```

Type the following command to stop the daemon process:

```
<workingdirectory>$ ddos -s | -stop
```